

## Opprett ordre

Rød (antall)

Svart (antall)

Testfase

Before = før optimalisering, After = etter optimalisering

Total: 10/12

### Vann % per cup

Bare cup 1..Total vises. Resten skjules automatisk.

Red 1

Red 2

Red 3

Red 4

Red 5

Black 1

Black 2

Black 3

Black 4

Black 5

Lagre ordre

Regel: Rød 0–6, Svart 0–6, Total 0–12.

Figur 1 Skjerm bilde av ERP-systemet, der operatøren kan opprette en ordre.

```
C:\Users\maxam\OneDrive\Di X + v

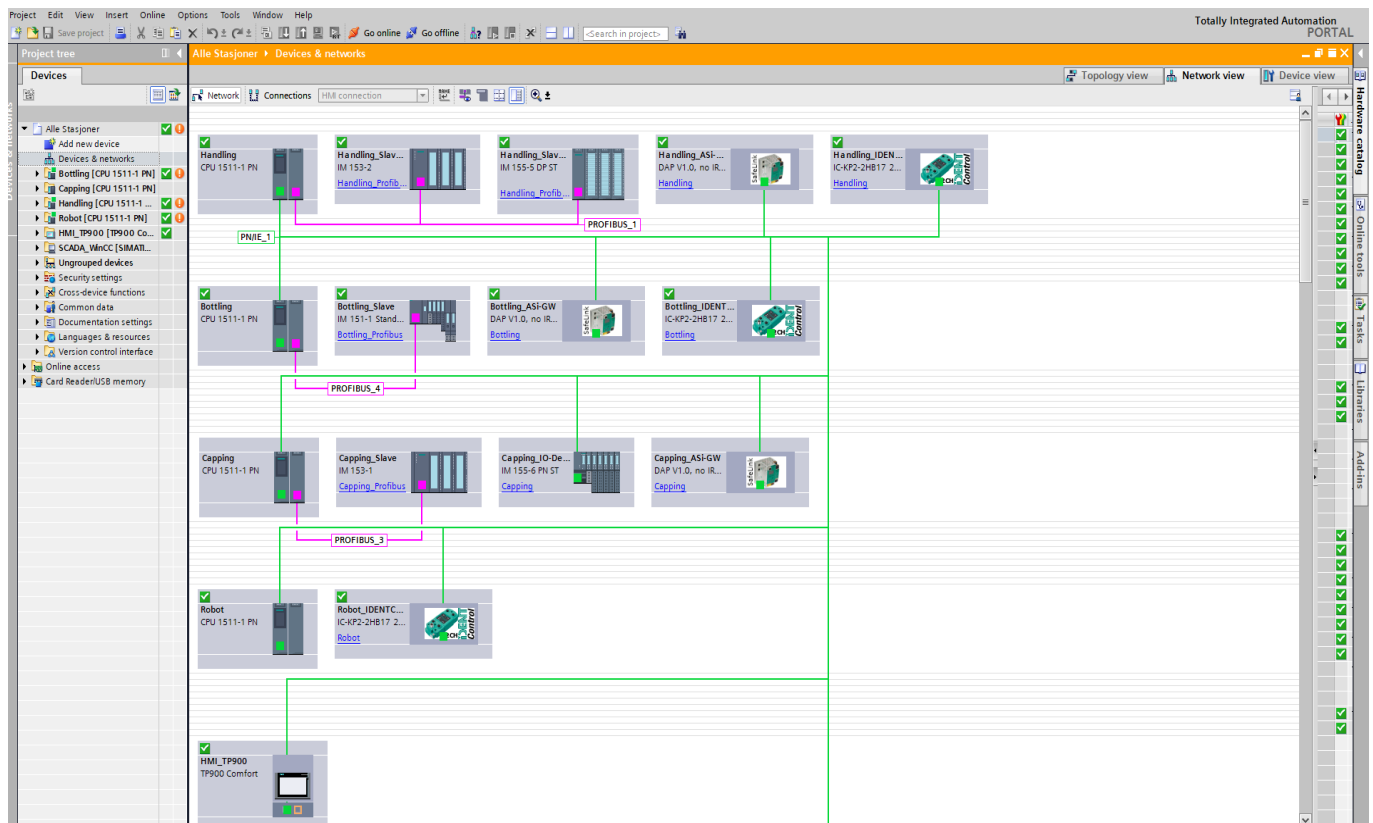
=== ERP Gateway ===
Handling PLC: 192.168.0.201
Robot PLC: 192.168.0.211
[11:10:45] Connecting PLC 192.168.0.211...
[11:10:45] PLC connected: 192.168.0.211
[11:10:45] OrderDone TRUE but no Running order found.
[11:13:15] Sending OrderId=22 to PLC... waiting for OrderAck=TRUE
[11:13:15] PLC ack received. SQL: Order 22 -> Ready
[11:16:30] SQL: Order 22 -> Running + StartedDate set
[11:16:30] Orders updated -> OrderId=22, QuantityProduced=0
[11:16:30] Orders updated -> OrderId=22, QuantityProductFault=0
[11:17:09] Orders updated -> OrderId=22, QuantityProductFault=1
[11:17:09] Robot QC saved -> OrderId=22, JarId=1, WeightDifference=-1,424, RealWeight=14,587, TimeSpentMinutes=0,02
[11:17:21] Orders updated -> OrderId=22, QuantityProductFault=2
[11:17:21] Robot QC saved -> OrderId=22, JarId=2, WeightDifference=-1,801, RealWeight=15,611, TimeSpentMinutes=0,1
[11:17:28] Orders updated -> OrderId=22, QuantityProductFault=3
[11:17:28] Robot QC saved -> OrderId=22, JarId=3, WeightDifference=-1,682, RealWeight=17,131, TimeSpentMinutes=0,12
[11:17:35] Orders updated -> OrderId=22, QuantityProductFault=4
[11:17:36] Robot QC saved -> OrderId=22, JarId=4, WeightDifference=-2,082, RealWeight=18,132, TimeSpentMinutes=0,12
[11:17:43] Orders updated -> OrderId=22, QuantityProductFault=5
[11:17:43] Robot QC saved -> OrderId=22, JarId=5, WeightDifference=-2,423, RealWeight=19,192, TimeSpentMinutes=0,1
[11:18:13] Orders updated -> OrderId=22, QuantityProductFault=6
[11:18:14] Robot QC saved -> OrderId=22, JarId=6, WeightDifference=-2,236, RealWeight=20,78, TimeSpentMinutes=0,5
[11:18:21] Orders updated -> OrderId=22, QuantityProductFault=7
[11:18:22] Robot QC saved -> OrderId=22, JarId=7, WeightDifference=-2,865, RealWeight=21,552, TimeSpentMinutes=0,5
[11:18:29] Orders updated -> OrderId=22, QuantityProduced=1
[11:18:30] Robot QC saved -> OrderId=22, JarId=8, WeightDifference=-0,812, RealWeight=25,006, TimeSpentMinutes=0,5
[11:18:38] Orders updated -> OrderId=22, QuantityProduced=2
[11:18:38] Robot QC saved -> OrderId=22, JarId=9, WeightDifference=-0,74, RealWeight=26,479, TimeSpentMinutes=0,5
[11:18:46] Order 22 marked for finish. Waiting 2 seconds for final PLC updates...
[11:18:46] Orders updated -> OrderId=22, QuantityProduced=3
[11:18:47] Robot QC saved -> OrderId=22, JarId=10, WeightDifference=0,415, RealWeight=29,035, TimeSpentMinutes=0,5
[11:18:48] SQL: Order 22 -> Finished + FinishedDate + TimeSpentMinutes (using actual OrderDone time)
```

Figur 2 Skjerm bilde av gateway-programmet.

|    |   |   |   |    |          |                         |                         |                         |    |   |      |        |
|----|---|---|---|----|----------|-------------------------|-------------------------|-------------------------|----|---|------|--------|
| 20 | 2 | 5 | 5 | 10 | Finished | 2026-04-14 11:02:40.790 | 2026-04-14 11:03:18.873 | 2026-04-14 11:05:44.000 | 10 | 0 | 2.42 | Before |
| 21 | 1 | 5 | 5 | 10 | Finished | 2026-04-14 10:53:42.143 | 2026-04-14 10:54:06.797 | 2026-04-14 10:57:07.460 | 10 | 0 | 3.01 | Before |

|     | CupId | OrderId | CupNo | Color | WaterPercent | WeightDifference | RealWeight | TimeSpentMinutes |
|-----|-------|---------|-------|-------|--------------|------------------|------------|------------------|
| 192 | 12    | 2       | 2     | Red   | 20           | 0.1797514        | 17.83225   | 0.72             |
| 193 | 13    | 2       | 3     | Red   | 30           | 0.1750813        | 19.23792   | 0.70             |
| 194 | 14    | 2       | 4     | Red   | 40           | 0.1123266        | 20.70167   | 0.68             |
| 195 | 15    | 2       | 5     | Red   | 50           | 0.003105164      | 22.21189   | 0.70             |
| 196 | 16    | 2       | 6     | Black | 60           | -0.1959257       | 23.42007   | 0.73             |
| 197 | 17    | 2       | 7     | Black | 70           | -0.0170002       | 25         | 0.72             |
| 198 | 18    | 2       | 8     | Black | 80           | -0.1052675       | 26.31273   | 0.70             |
| 199 | 19    | 2       | 9     | Black | 90           | 0.5383415        | 28.35734   | 0.72             |
| 200 | 20    | 2       | 10    | Black | 100          | 0.2409658        | 29.46097   | 0.73             |

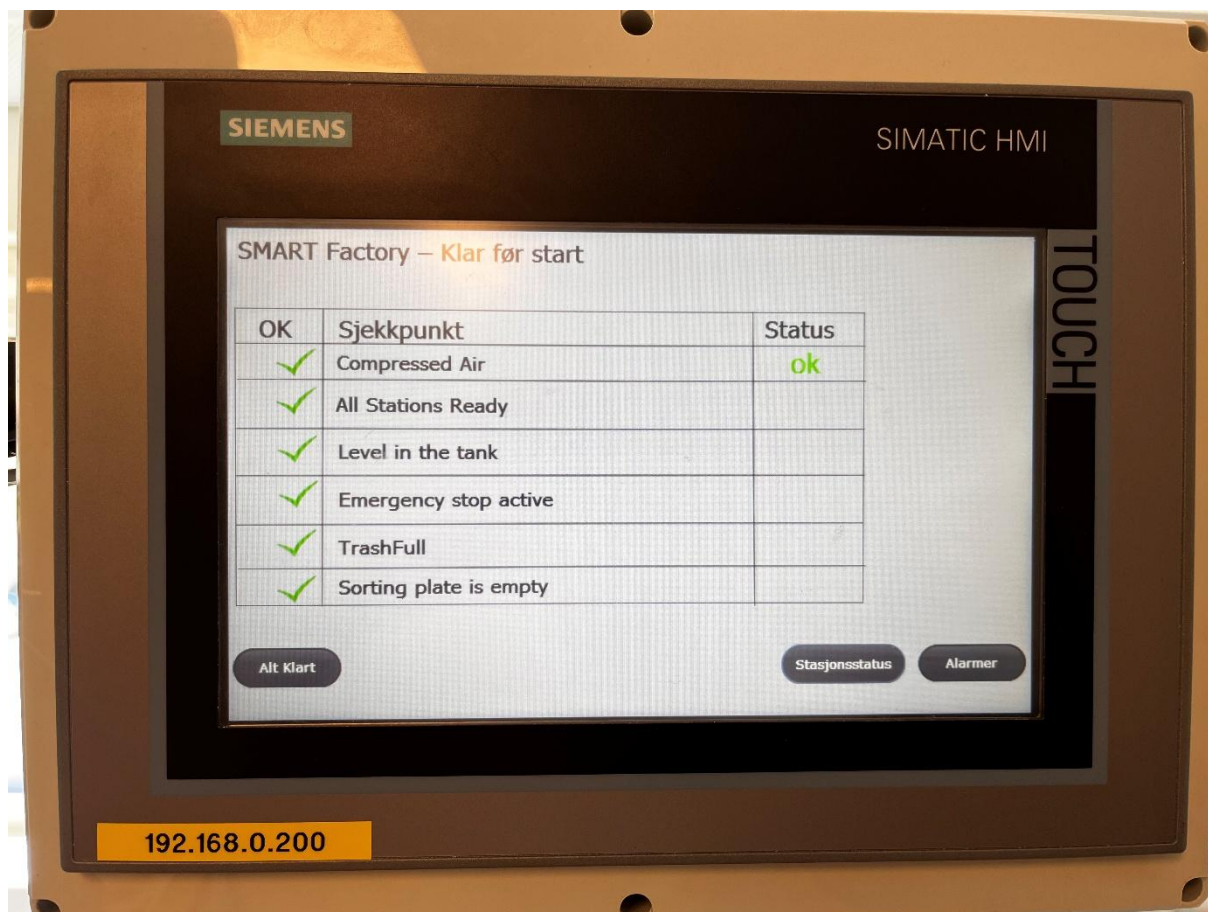
Figur 3 Skjerm bilde av SQL-databasen.



Figur 4 Skjerm bilde fra TIA Portal V17 med oversikt over alle komponenter.



*Figur 5 Bilde av enkoder på handling-stasjonen.*



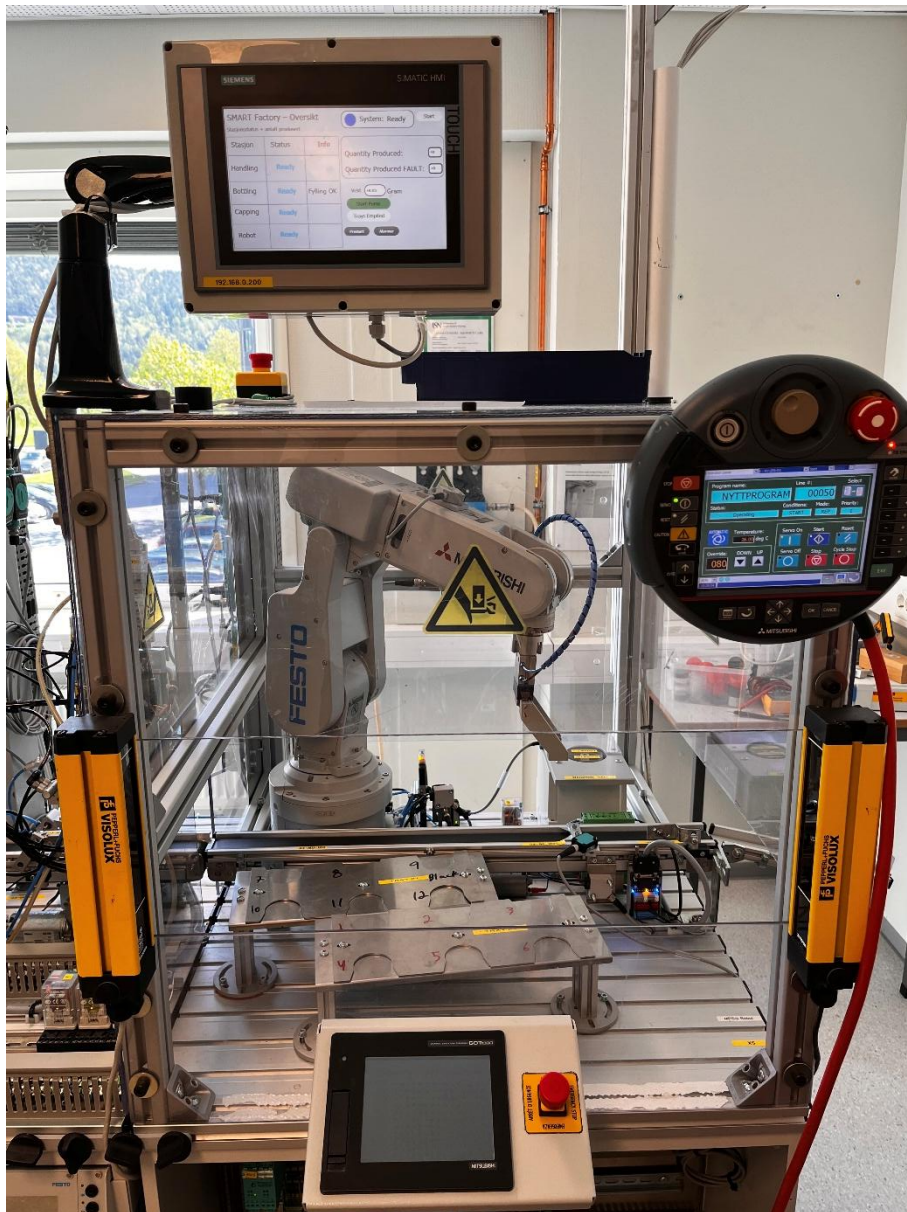
Figur 6 HMI som viser at anlegget er klart for start.



Figur 7 Robotens fjernkontroll.



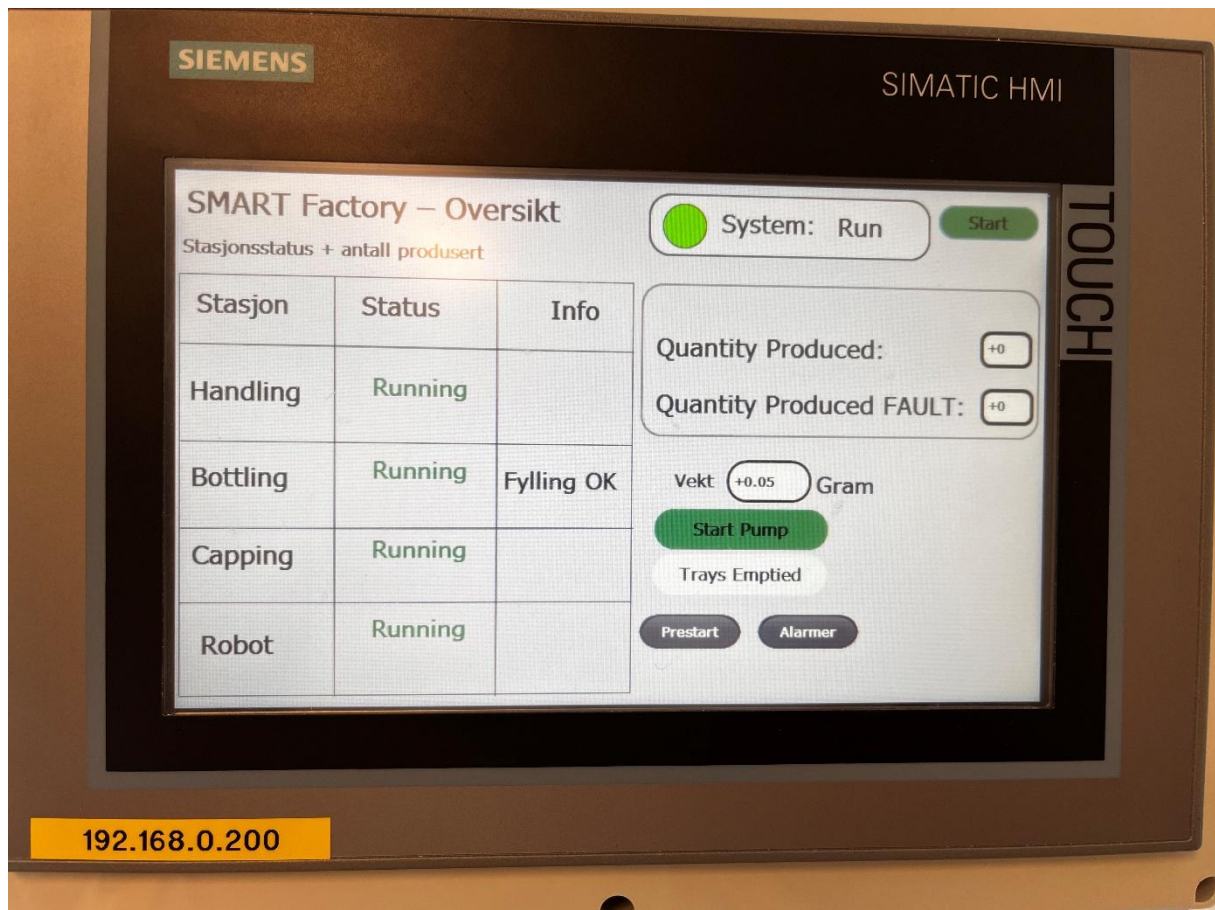
Figur 8 Mitsubishi robotkontroller CR750-02-VD1-1-S15.



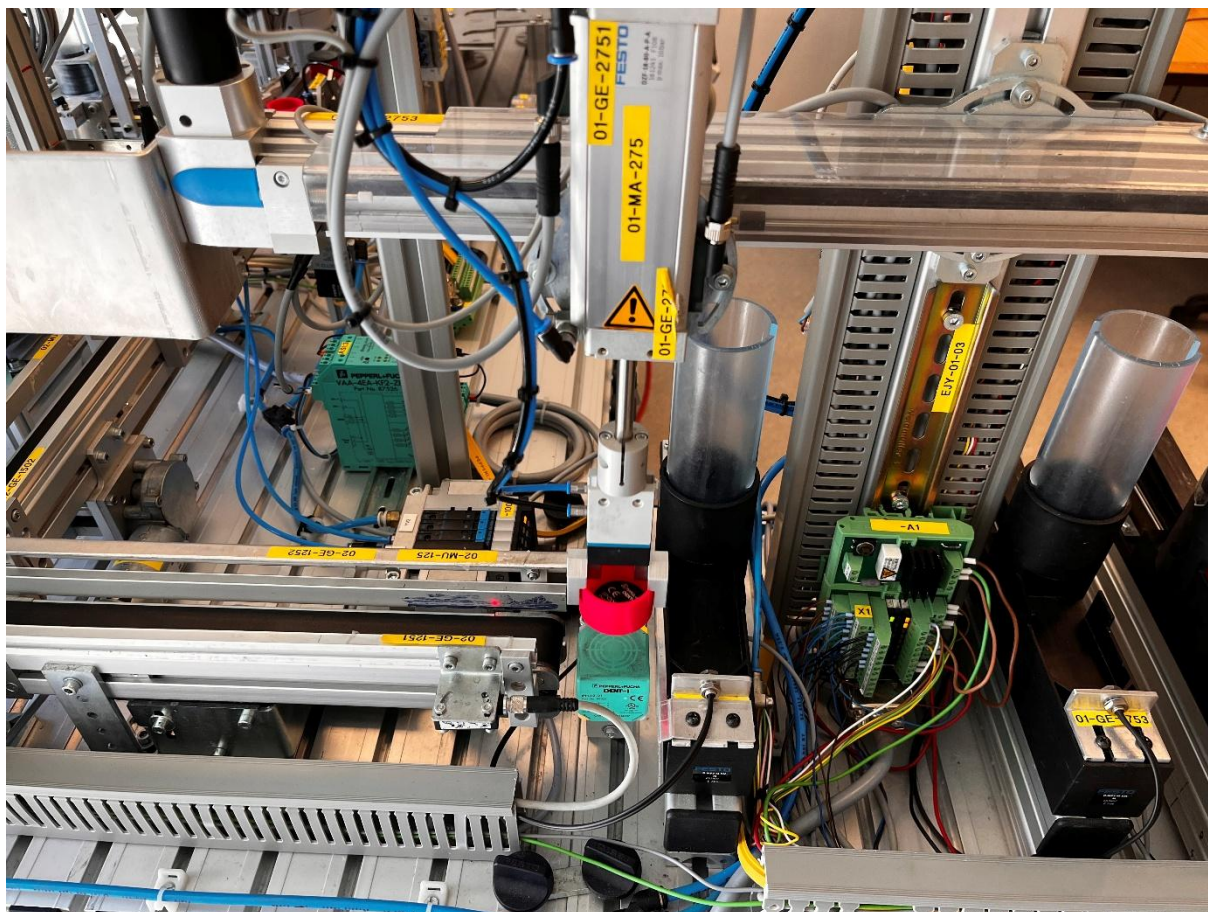
Figur 9 Robot Stasjon.



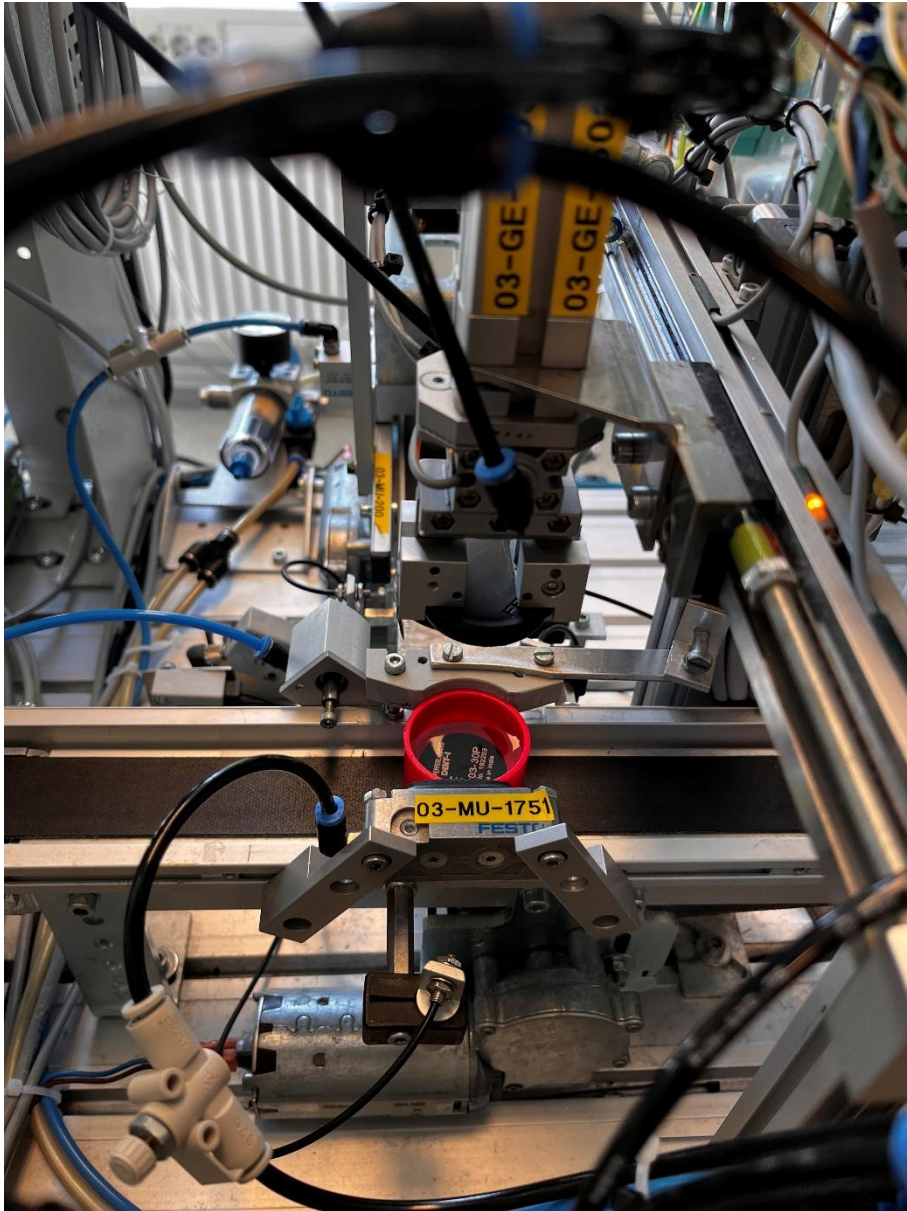
Figur 10 HMI-oversikt



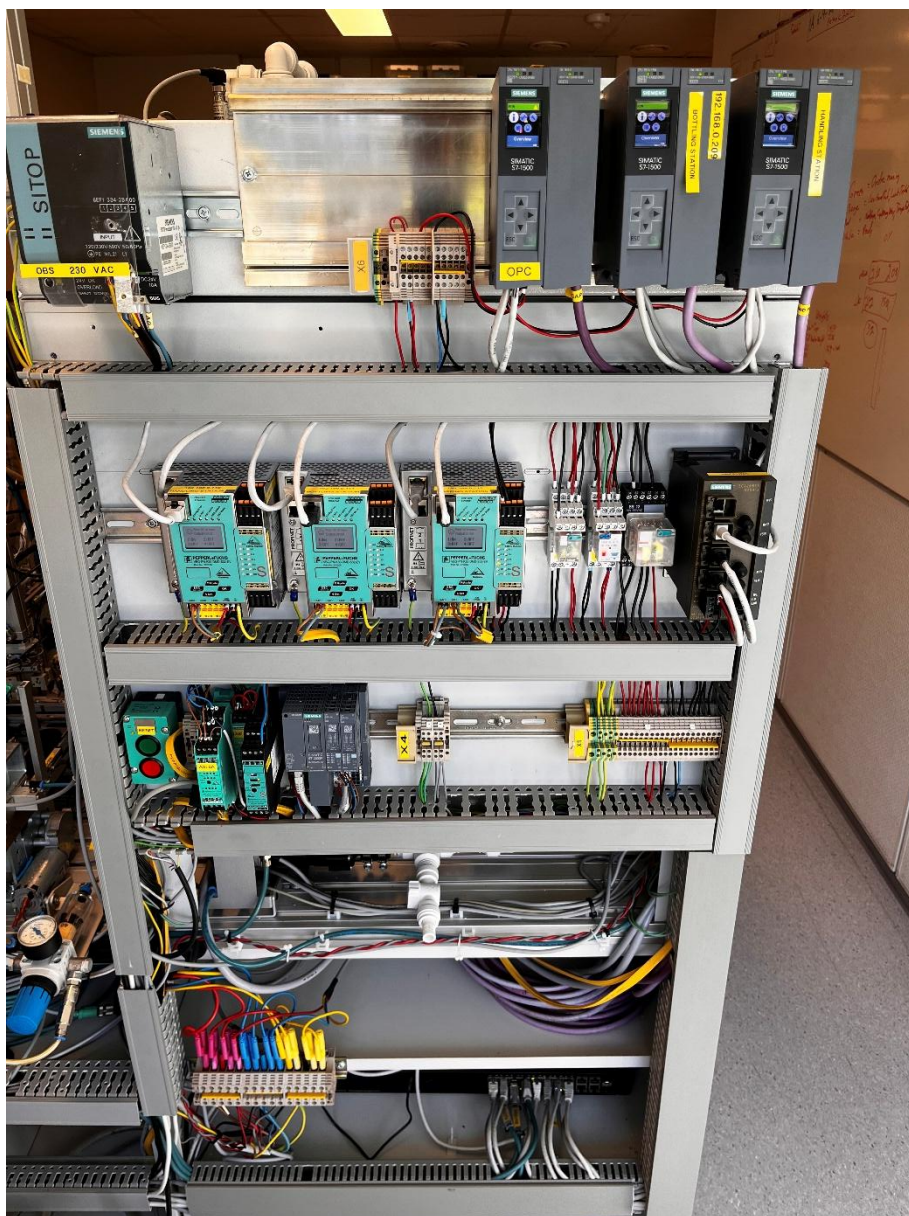
Figur 11 HMI som viser Festo-anlegget i drift.



Figur 12 Kopper ved RFID-leser på handling-stasjonen.



Figur 13 Kopp etter lukking.



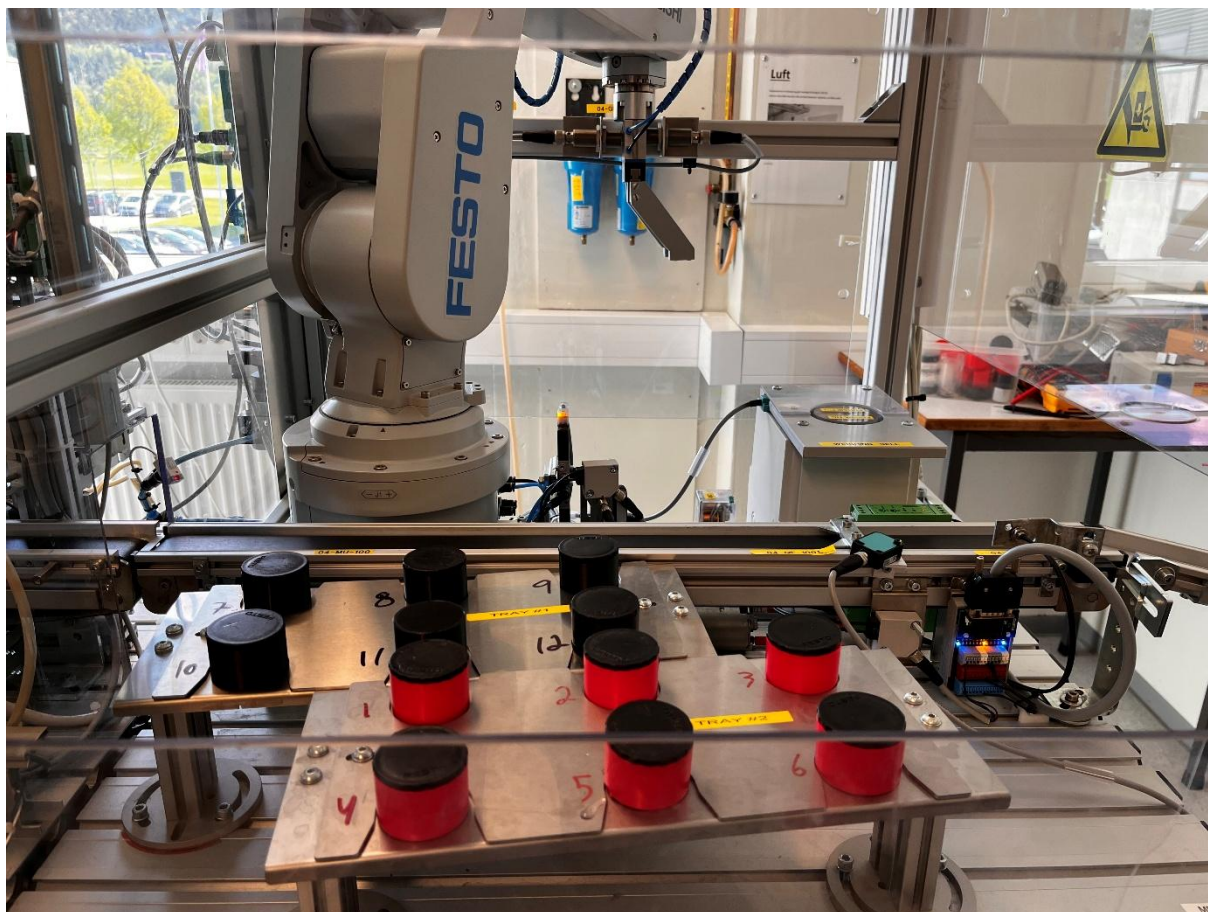
Figur 14 Bilde av PLS S7-1500, ASI Gateway, Safe link og Nettverk Switch helt ned



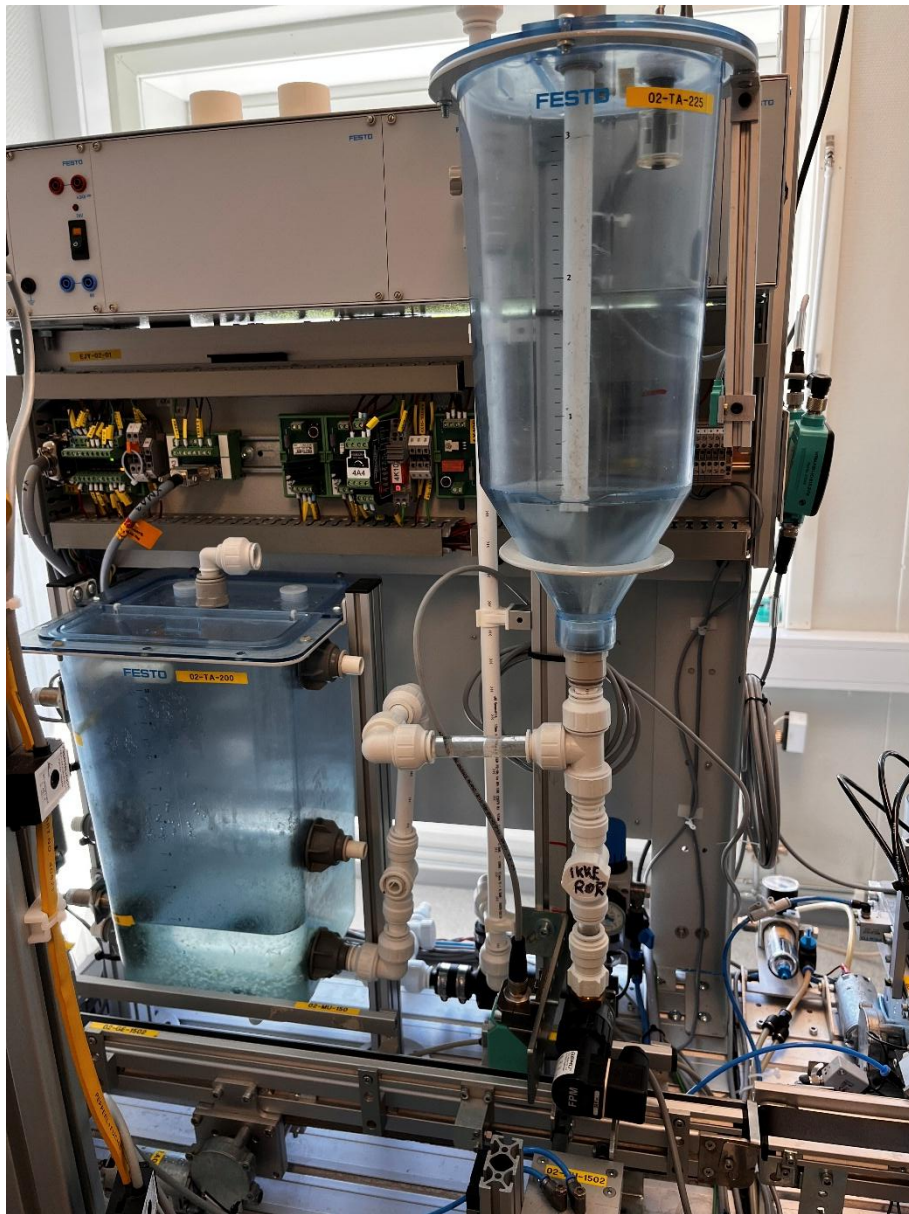
Figur 15 Nødstopp på bottling-stasjonen med reset-knapp



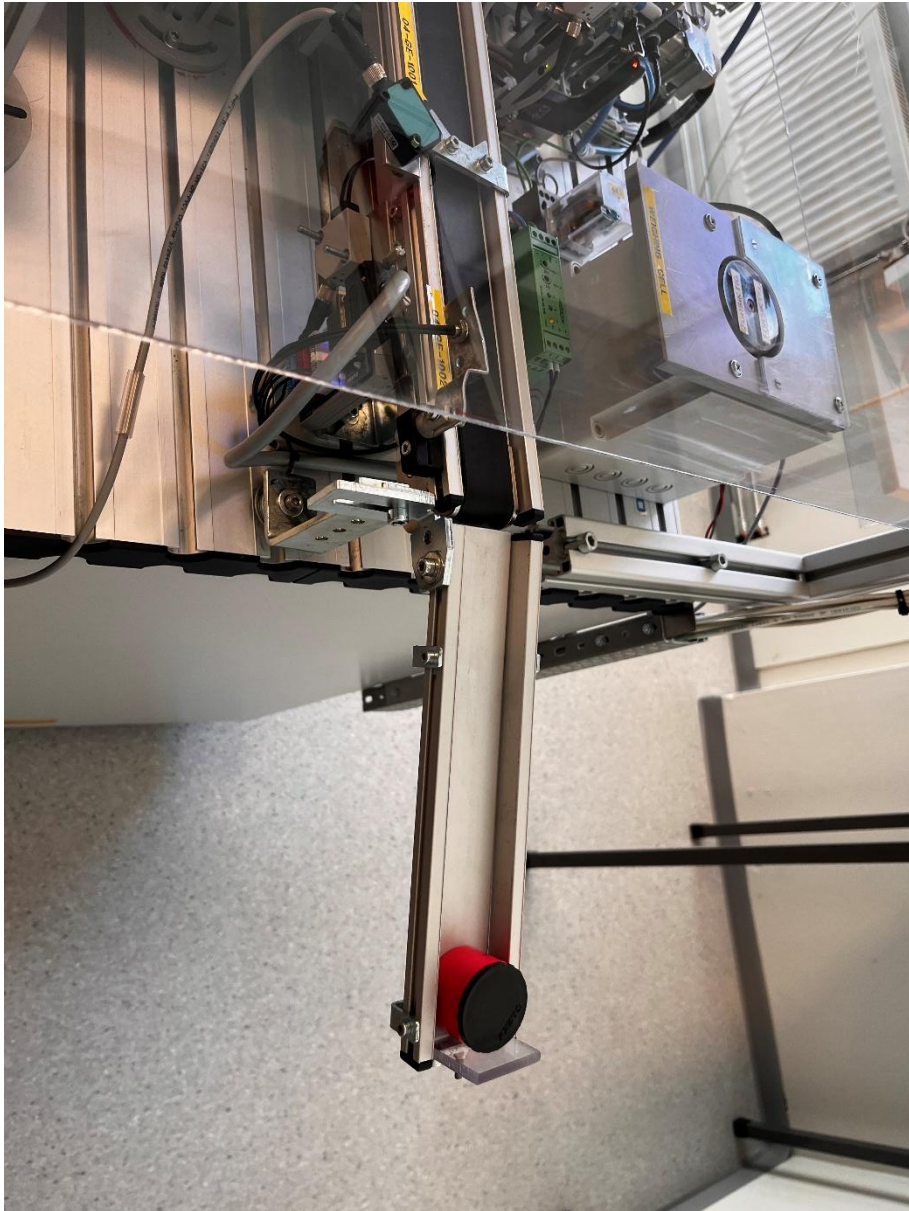
*Figur 16 Robot i sikker posisjon. Operatøren kan da føre hånden inn i robotstasjonen uten at lysgardinen utløser nødstop.*



Figur 17 : Ferdig ordre med roboten i sikker posisjon.



Figur 18 Tank til fylling.



Figur 19 Kopper som ikke er godkjent.



*Figur 20 Luftkompressor.*



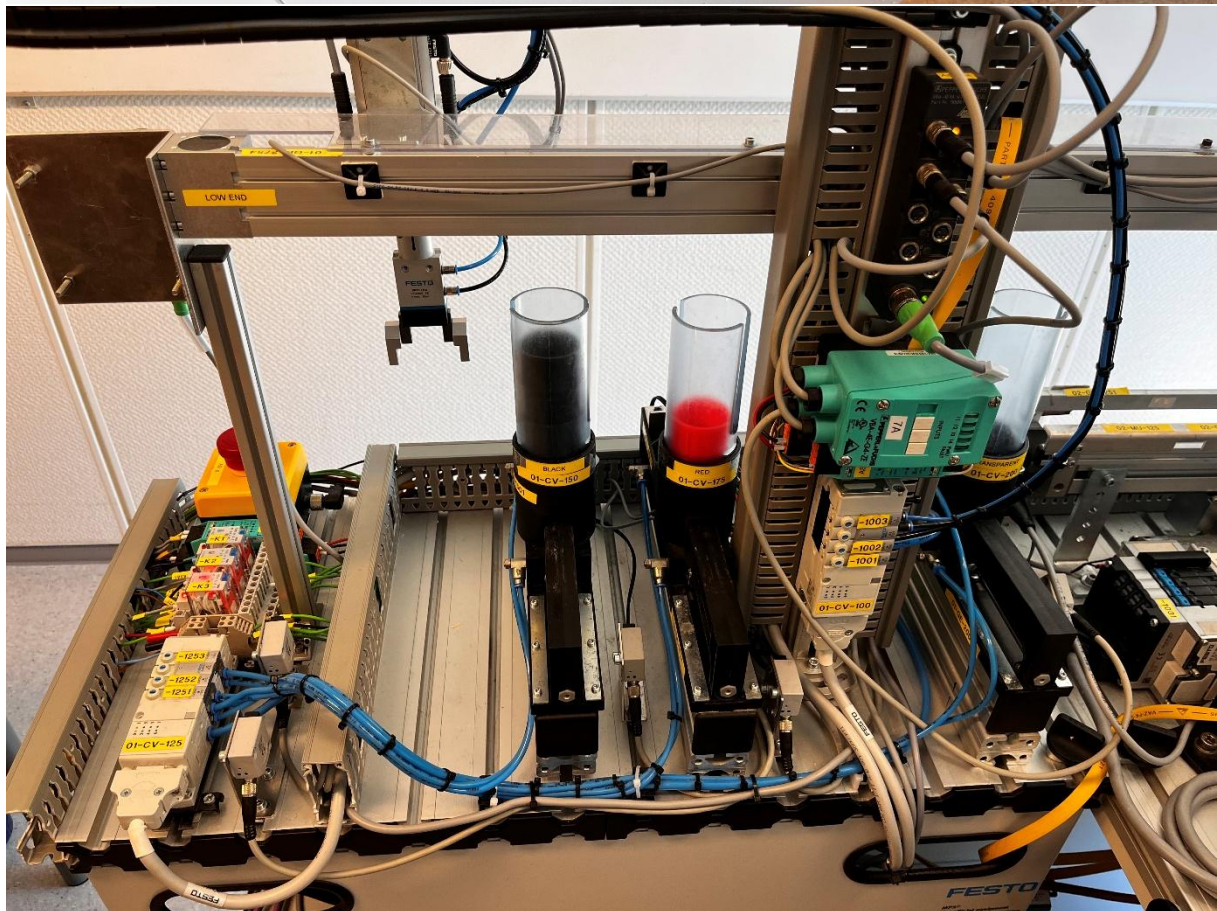
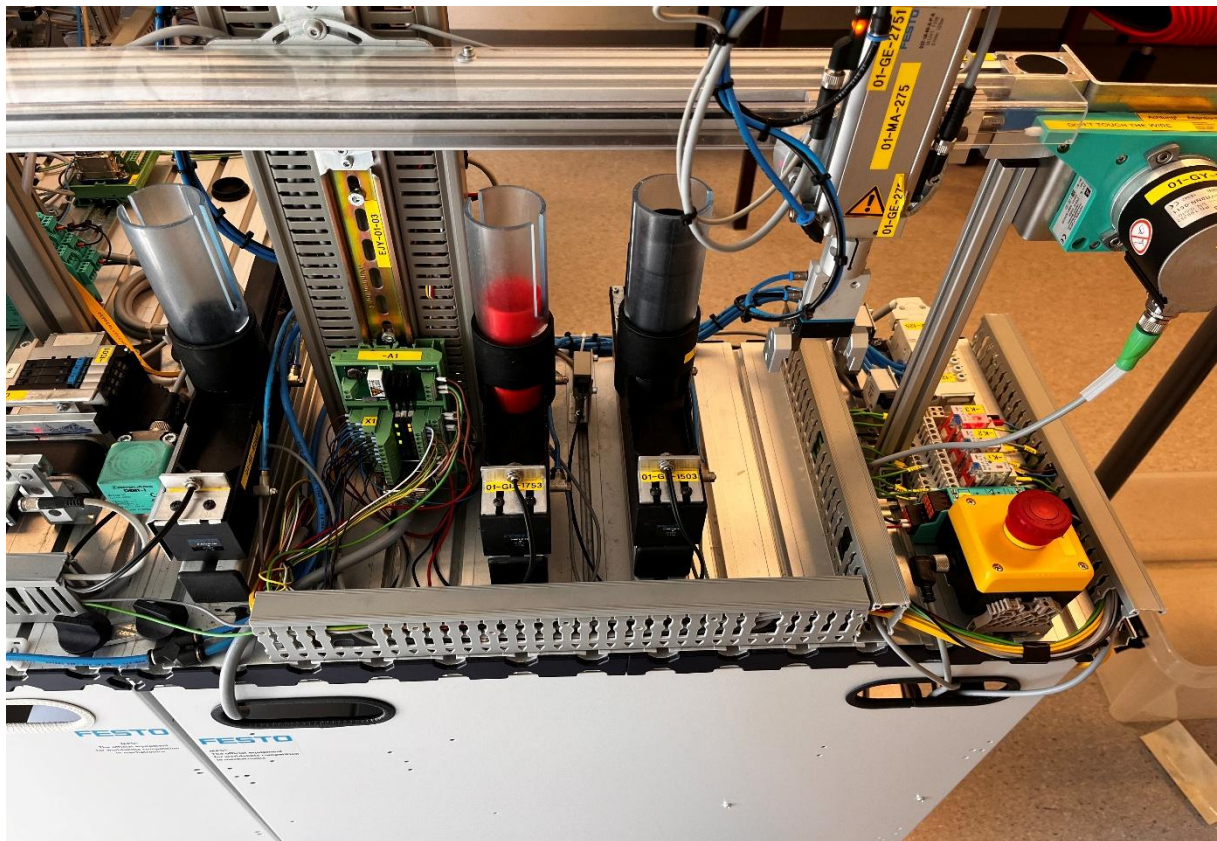
Figur 21 Vekt/veiecelle for veiing, RFID-lesing og i robotstasjonen.



*Figur 22 Robot stasjon fra bak*



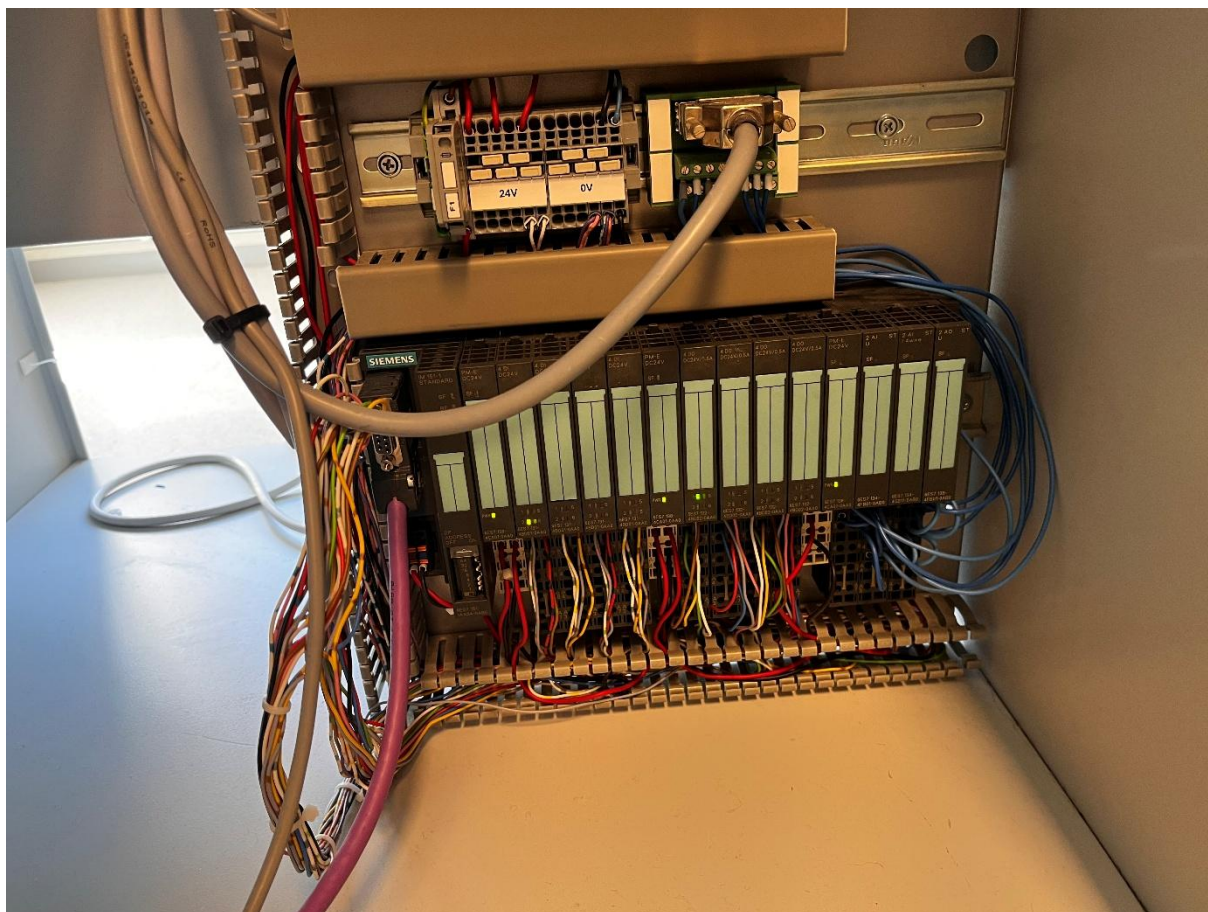
Figur 23 Robot-PLS S7-1500 for tilbakemelding og robotens Interface-boks 534364.



Figur 24 Handling-stasjon



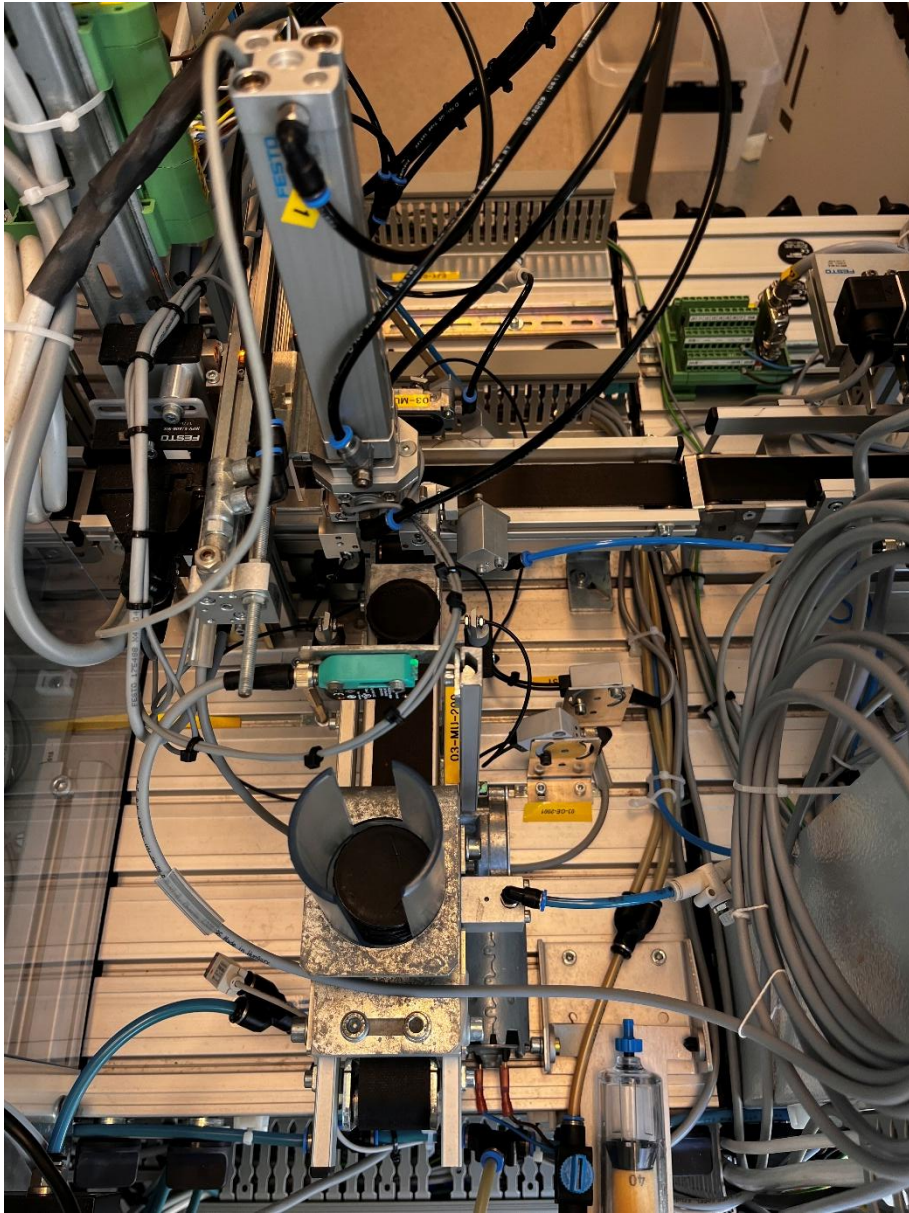
*Figur 25 Varsellys.*



Figur 26 I/O-moduler til fylling-PLS.



Figur 27 RFID-brikker.



Figur 28 Capping-stasjon.